ions provide a reflecting medium for the reflection of radio waves or communication waves back to Earth within the permitted frequency range. The phenomenon of bending the radio waves back to Earth is nothing but the total internal reflection.

When the angle of incidence at the ionosphere is large, the sky wave returns to the ground at a long distance from the transmitter. As this angle is reduced, the wave returns closer and closer to the transmitter. If the angle of incidence is reduced further, the radio waves penetrate through the ionosphere. For a particular angle of incidence, the point of reception (B) is at the minimum distance from transmitter. **The shortest distance between the transmitter and the point of reception of the sky wave along the surface is called as the skip distance**.

The ground waves get attenuated as they move away from the transmitter. At a particular point (A), there is no reception ground wave. **The zone (in between A and B) where there is no reception of electromagnetic waves neither ground nor sky is known as skip zone or skip area** (Figure 10.53(b)).

iii) Space wave propagation

The process of sending and receiving information signal through space is called space wave communication (Figure 10.53(c)). **The electromagnetic waves of very high frequencies above 30 MHz are called as space waves.** These waves travel in a straight line from the transmitter to the receiver. Hence, it is used for a line of sight communication (LOS).

The communication systems like television telecast, satellite communication and RADAR are based on space wave propagation.

The range or distance (d) of coverage of the propagation depends on the height (h) of the antenna given by the equation,

$$d = \sqrt{2Rh} \qquad (10.9)$$

where R is the radius of the Earth. The distance of coverage is shown pictorially in Figure 10.54.

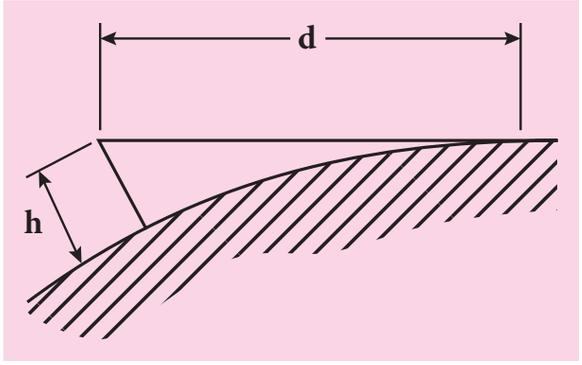
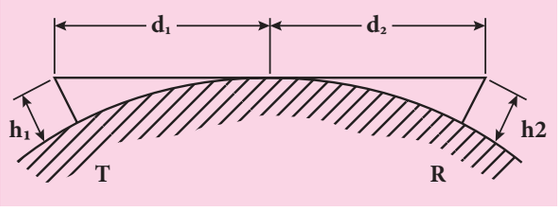


Figure 10.54 Distance of coverage

EXAMPLE 10.12

A transmitting antenna has a height of 40 m and the height of the receiving antenna is 30 m. What is the maximum distance between them for line-of-sight communication? The radius of the earth is 6.4×10^6 m.



Solution:

The total distance *d* between the transmitting and receiving antennas will be the sum of the individual distances of coverage.

XII_Physics_Vol-2_UNIT-10_Part 2_EM.indd 239 04-03-2022 17:19:22